ECON 0100 | Demo D

Demos are similar to Checkpoints, often taken directly from past semesters. Work through the problems and check your work against mine. Practice answering clearly and completely. You are encouraged to work in small groups. Find a study room, grab some classmates, and work together on a whiteboard.

Hermione's Potion Ingredients

Hermione and Draco share a garden plot growing moonflowers for their potions project. Each can harvest 1 or 2 flowers weekly. More harvesting reduces future growth. The payoff matrix shows their utility from different harvesting choices.

		Draco H1	**Draco** H2
Hermione	Harvest 1 (H1)	10, 10	6, 15
	Harvest 2 (H2)	14, 6	8, 8

Outcomes are written (Hermione's choice, Draco's choice).

D1.1 | Best Responses

	Hermione	**Draco**
Best Response When the Other Harvests 1:	_____	_____
Best Response When the Other Harvests 2:	_____	_____
Dominant Strategy:	☐ H1 ☐ H2 ☐ none	☐ H1 ☐ H2 ☐ none

D1.2 | Nash Equilibrium & Efficiency

Nash Equilibrium (*check all that apply*):	☐ (H1, H1)	☐ (H1, H2)	☐ (H2, H1)	☐ (H2, H2)
Socially Efficient Outcome (*check all that apply*):	☐ (H1, H1)	☐ (H1, H2)	☐ (H2, H1)	☐ (H2, H2)
Total Payoff, Nash Equilibrium / Efficient Outcome:	_____ / _____			
An Outcome Both Prefer to the Nash Equilibrium:	☐ (H1, H1)	☐ (H1, H2)	☐ (H2, H1)	☐ none

D2.1 | Common Resources

The Moonflowers in the Plot Are: ☐ rival ☐ non-rival ☐ excludable ☐ non-excludable

If Hermione Switches From (H1, H2) to (H2, H2): Hermione: + _____ Draco: − _____

Professor Sprout begins checking the plot each week and fines anyone who harvests 2 flowers. The fine lowers that student's payoff by 6. Fill in the payoff matrix with the fine.

Nash Equilibrium With the Fine:
☐ (H1, H1) ☐ (H1, H2)
☐ (H2, H1) ☐ (H2, H2)

		Draco H1	**Draco** H2
Hermione	H1		
	H2		

Moonflower Management

In this question we will consider whether to impose a harvesting restriction. Five prefects harvest moonflowers from the Forbidden Forest. Without restrictions, the flowers will go extinct after this year. The table shows each prefect's annual benefits (in coins), both under current harvesting and with proposed restrictions that would prevent extinction. The restrictions cost 60 coins yearly, split equally (12 coins each).

Benefits (*in coins*)	**Hermione**	**Draco**	**Hannah**	**Cho**	**Padma**
Without restrictions	25	22	18	16	12
With restrictions (*not including fee*)	40	35	32	20	15

D3.1 | Public Goods

Keeping the Moonflowers From Extinction Is: ☐ rival ☐ non-rival
☐ excludable ☐ non-excludable

Prefects Who Would Pay Voluntarily, With No Fee: _____

Marginal Social Benefit of the Restrictions: _____

Providing the Restrictions Is: ☐ efficient ☐ not efficient

D3.2 | Voting on Public Goods

Hermione's Benefit Without Restrictions: _____

Hermione's Benefit With Restrictions (*after the fee*): _____

Hermione's Vote: ☐ for ☐ against

Prefects Voting For / Against: _____ / _____

Total Benefit Without Restrictions: _____

Total Benefit With Restrictions (*after the fees*): _____

Restrictions With a 12-Coin Fee: ☐ efficient ☐ not efficient